

# Patrick Manning Kennedy

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## EDUCATION

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### Embry-Riddle Aeronautical University

Daytona, FL

Major: Aerospace Engineering, Minor: Computer Science, GPA: 3.943

Dec 2024

## EXPERIENCE

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### ASUS Robotics & AI Center

Taipei City, Taiwan

*Systems Engineer*

Jul 2025 — Present

- Architected a distributed, asynchronous messaging framework using Rust and C++ for dynamic autonomous systems.

*Computer Vision Engineer*

Feb 2025 — Jul 2025

- Researched, designed and implemented an efficient VIO system for real-time localization and control.

### NASA funded Swarm Unmanned Aerial Vehicle using Emergence (SUAVE), EPPL

Daytona, FL

*Software Lead*

Jan 2024 — Dec 2024

- Utilized \$86K budget to design and implement a GPS-denied UAV swarm for exploration and mapping.
- Research computer vision techniques including Visual-SLAM, HSV masking, and Collaborative SLAM.
- Computed local odometry with V-SLAM to enable GPS-denied flight using ROS2 and MAVLink in C++.
- Independently designed a robust drone controller interface in C++ with an emphasis on safety and portability.
- Implemented and tuned 4 PID controllers for a constrained follow me' UAV position control.
- Developing a dual quaternion swarm controller that emulates dynamics and kinematics of a virtual structure to navigate a swarm of drones from one pose to another.

### Guave Compiler

*Independent Project*

Jun 2020 — May 2023

- Developing a statically typed language in C++ intended to generate safe and modular code.
- Designed a dynamic grammar focusing on readability while promoting modern design patterns and practices.
- Implementing macros in native Guava syntax to modify the intermediate representation (IR) on compile time.
- Exposing Guava compiler plugin points for custom runtime targets such as LLVM IR, transpiled GPGPU code, or interpretation, while maintaining interoperability between targets.

### Ansys Government Initiatives

Exton, PA

*STK Communications & Radar Developer*

May 2023 — Aug 2024

- Designed and developed complex radar workflows in C++ including 3D antenna gain pattern graphics, and real-time SAR radar image computation and overlay.
- Researched and interfaced with propagation models including GPU accelerated shooting and bouncing rays.
- Utilized COM and ATL in C++ and C# for interoperability between different programs and API generation.
- Implemented gRPC to create a thread safe server client interface into STK, enabling a scenario dependent controller to be executed in parallel to a real-time simulation.

*STK Quality Assurance Intern*

Sept 2022 — Dec 2022

- Conducted thorough vulnerability analysis of the 10-dimensional MODTRAN-derived model in Python.
- Developed regression test in Perl flagging interpolation inaccuracies within the MODTRAN-derived model.
- Evaluated EOIR Sensor API feature to identify release critical bugs or unexpected behaviors.
- Developed regression test in Perl ensuring functionality is retained for future releases across all edge cases.

## PUBLICATIONS

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D. Golan, **P.Kennedy**, B. Gonzalez, et al., “Swarm UAVs for area mapping in GPS-denied locations”, SPIE Defense and Commercial Sensing Conference Proceedings, Paper No. 13055-28, June 2024.